YEAR 7 · SCIENCE 3.1 · LAB

Drop It!

How fast are you? Do heavy things fall faster?

Name: _____ **Class:** _____ **Date:** _____

My group: 1. _____ 2. _____ 3. _____

Predict first — on your own

Mr Bowen drops a heavy ball and a light ball, the same size, at the same time. Which lands first? ☐ heavy ☐ light ☐ same time

I think … because …

_____

Stay safe

- Only drop objects **Mr Bowen has checked**.
- **Let go. Never throw. Never catch.**
- Part 2: feet on the floor. No chairs or desks.
- Part 3: stay **behind the railing**. Never lean.
- Part 3: drop only after the Ground person shouts **"Clear!"**

Two tools: you can use a calculator

Mean = add up all the values, then divide by how many there are.

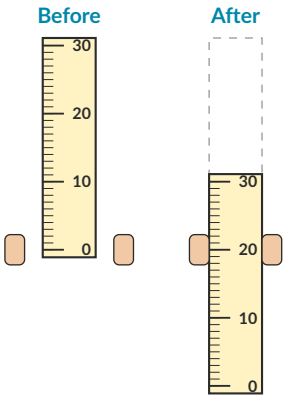
Speed = distance ÷ time, in **m/s** (**metres per second**). **cm to m:** ÷ 100

Mr Bowen: 20 cm = 0.20 m, in 0.20 s, so speed = 0.20 ÷ 0.20 = **1.0 m/s**.

Part 1 · The Ruler Catch — How Fast Are You?

Jobs: Catcher | Dropper | Timer

1. **Catcher:** rest your arm on the desk. Thumb and finger open, at the **0**.
2. **Dropper:** hold the top. Let go with **no countdown**!
3. **Catcher:** catch it. Read the cm at the **top of your thumb**.
4. **Timer:** try to time the fall with a stopwatch.
5. **5 catches each.** Then change jobs.



20 cm = 0.20 s

1A The ruler table: where you caught it → your reaction time

Caught at (cm)	1	2	3	4	5	6	7	8	9	10
Time (s)	0.04	0.06	0.08	0.09	0.10	0.11	0.12	0.13	0.13	0.14
Caught at (cm)	11	12	13	14	15	16	17	18	19	20
Time (s)	0.15	0.15	0.16	0.17	0.17	0.18	0.18	0.19	0.19	0.20
Caught at (cm)	21	22	23	24	25	26	27	28	29	30
Time (s)	0.20	0.21	0.21	0.22	0.22	0.23	0.23	0.24	0.24	0.24

1B My five catches

Catch	1	2	3	4	5	Total	Mean (÷ 5)
Caught at (cm)							
Ruler time (s)							
Stopwatch time (s)							

Speed of the ruler = mean catch (m) ÷ mean time = _____ ÷ _____ = _____ **m/s**

Think Your ruler times and stopwatch times are different. Why is a stopwatch no good for timing the ruler? Use **reaction time**.
The fall only takes about … The Timer …

Part 2 · Two Objects in the Lab — Do Heavy Things Fall Faster?

Our objects, the **same size and shape**: **A** (heavy) _____ **B** (light) _____

Jobs: Dropper | Timer A | Timer B Checked by Mr Bowen ☐

1. Put tape on the wall, as high as the **shortest** person can reach. Drop from this mark **every time**.
2. Dropper: hold A and B at the tape. Say **"3, 2, 1, drop!"** and let go of both together.
3. Timer A times A. Timer B times B. Everyone listens: **one bang or two?**
4. Do **10 drops**. Change jobs after drop 3 and drop 6.

Height of the tape: _____ cm = _____ m

2A Ten drops

Write each time like this: 0.63. Last row: who landed first? **A**, **B** or **S** (same time).

Drop	1	2	3	4	5	6	7	8	9	10	Total	Mean
A, heavy (s)												
B, light (s)												
First? A, B, S												

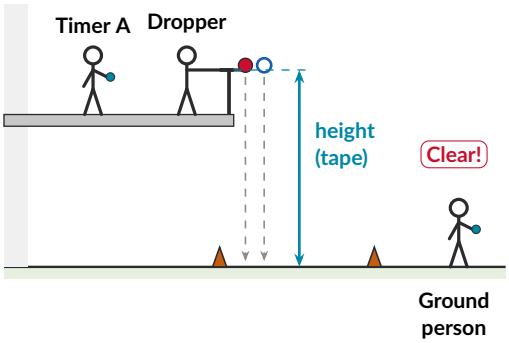
Speed of A = height (m) ÷ mean time = _____ ÷ _____ = _____ **m/s**

Think Why did we **change** Timers? Why must A and B be the same size and shape?
So that a slow Timer … If one is bigger or flatter, the air …

Part 3 · From the Balcony — How Fast Does It Go?

Upstairs: Dropper Timer A
Downstairs: Ground person, who also times B

1. **Height:** the Ground person holds the **0** of the long tape on the ground. Upstairs, read it at the **top of the railing**. Three readings.
2. The Ground person stands well back and shouts **"Clear!"**
3. The Dropper holds A and B out over the railing, level with the top: **"3, 2, 1, drop!"**
4. Timer A times A. The Ground person times B, then **walks** them back up.
5. Change jobs after every drop. Do **6 drops**.



3A Height and prediction

	Reading 1	Reading 2	Reading 3	Mean (÷ 3)
Height (m)				

How long a fall takes with **no air**:

Height (m)	1.8	2.0	2.2	3.0	3.5	4.0	4.5	5.0	5.5	6.0	7.0
Time (s)	0.60	0.63	0.66	0.77	0.84	0.89	0.95	1.00	1.05	1.10	1.18

Predict. From our height, the fall takes about _____ s.

Which lands first from up here? ☐ A ☐ B ☐ same time

3B Six drops from the balcony

Drop	1	2	3	4	5	6	Total	Mean (÷ 6)
A, heavy (s)								
B, light (s)								
First? A, B or S								

Speed of A = height (m) ÷ mean time = _____ ÷ _____ = _____ **m/s**

Think Your reaction time is about 0.2 s. Which part gave the more **accurate** time, Part 2 or Part 3? Why?
The fall in Part … was longer, so 0.2 s is …

Reflection · Dear Aristotle

More than 2000 years ago, **Aristotle** said that heavy things fall faster. People believed him for about 2000 years. Write him a polite letter.

Your letter in four parts

Claim	your answer. Heavy things …
Evidence	at least **two numbers** from your tables. In the lab, the mean times were … From the balcony, …
Reasoning	the science. Gravity pulls the heavy object harder, but it is also harder to speed up, so …
Next time	one way to make the test more accurate. Next time we could …

Dear Aristotle,

From, _____